Time to Completion for Corporate Takeovers

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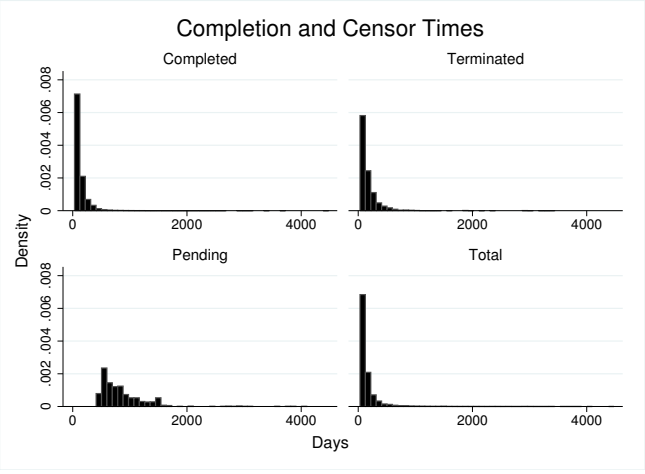
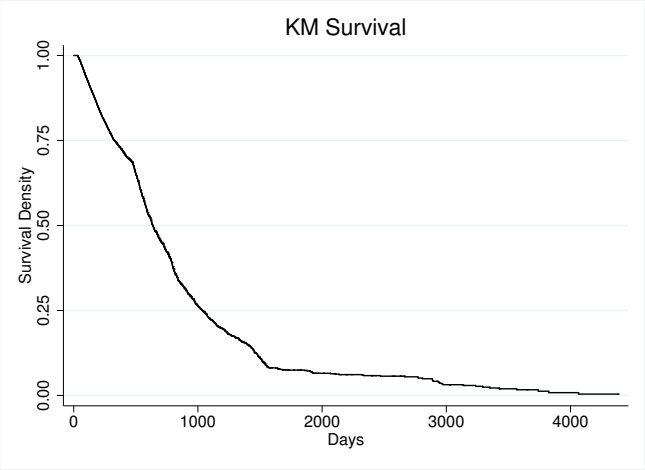
Time to Completion

- Risk inherent in down time for transactions
  - Target's management
  - Lost synergies
- What affects the time spent closing a deal???
  - The deal itself
  - The companies involved
  - The exogenous environment

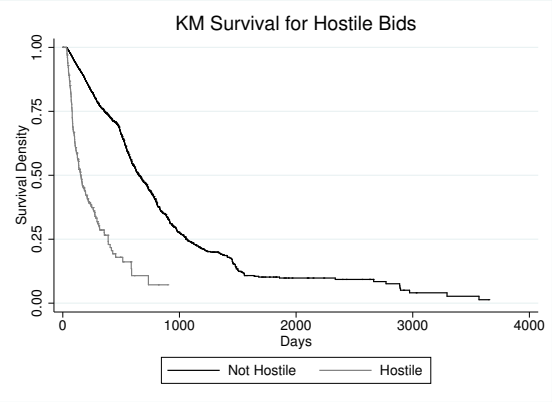
Data and Methodology

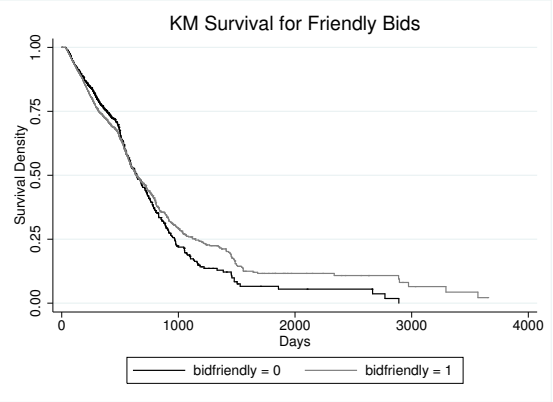
- Data: 46,902 Deals spanning 2000-2010
- Analysis uses the following:
  - Kaplan-Meier Estimator
  - Cox Regression
    - ★ $\lambda(t) = \lambda_0(t)e^{\beta_1 x_1 + \beta_2 x_2 + \ldots}$
  - Collett's Model Selection Approach
    - ★ Forward testing
    - ★ Backward testing
    - ★ Stepwise testing

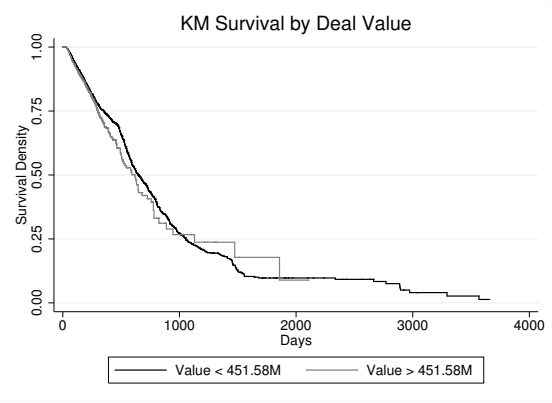


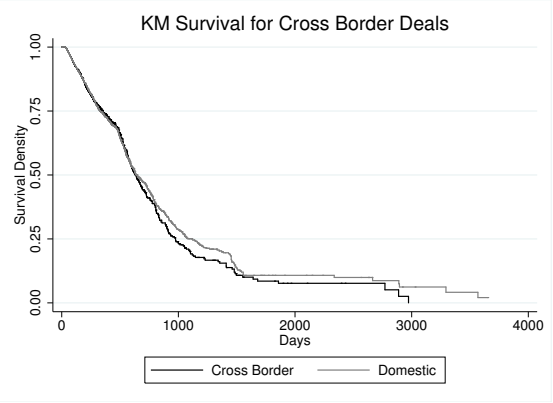


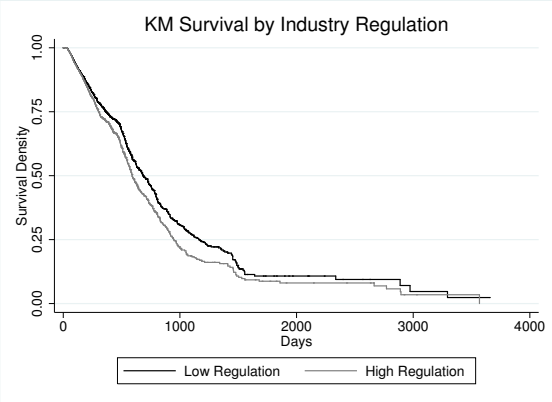
Deal Effects











Forward Regression, $n = 30{,}906$		
Covariate	β_i	Z Statistic
Hostile Bid	2.0549	25.97^{***}
Friendly Bid	0.3735	7.91^{***}
Industry Regulation	0.1555	4.42^{***}
Deal Value	-8.85e-6	-1.90^{*}
Backward Regression, $n = 30{,}906$		
Covariate	β_i	Z Statistic
Hostile Bid	2.0549	25.97^{***}
Friendly Bid	0.3735	7.91^{***}
Industry Regulation	0.1555	4.42^{***}
Deal Value	-8.85e-6	-1.90^{*}
Stepwise Regression, $n = 30{,}906$		
Covariate	β_i	Z Statistic
Hostile Bid	2.0549	25.97^{***}
Friendly Bid	0.3735	7.91^{***}
Industry Regulation	0.1555	4.42^{***}
Deal Value	-8.85e-6	-1.90^{*}

Company and Country Effects

Company Effects

Stepwise Regression, $n = 7{,}010$			
Covariate	β_i	Standard Error	Z Statistic
Acquirer Profit/Loss	0.3463	0.0798	4.34***
Target Assets	7.98e-6	1.86e-6	4.29***
Acquirer Assets	-1.27e-6	5.31e-7	-2.39**
Target Profit/Loss	0.1643	0.0757	2.17**

Country Effects

Stepwise Regression, $n = 31{,}777$			
Covariate	β_i	Standard Error	Z Statistic
Acquiring Country Corruption	0.3006	0.0901	3.34***
Difference in Corruption	-0.2824	0.0324	-8.71***
Acquiring Country Accountability	-0.2772	0.0425	-6.51***
Acquiring Country Stability	-0.1716	0.0483	-3.55***
Acquiring Country Rule of Law	0.3719	0.1209	3.08***

Final Results

Combined Model

Stepwise Regression, $n = 6{,}421$				
Covariate	β_i	Standard Error	Z Statistic	Deal Length
Hostile Bid	1.2067	0.1709	7.06^{***}	-
Friendly Bid	-0.3922	0.1526	-2.57^{**}	+
Industry Regulation	0.1284	0.0755	1.70^{*}	More: -
Acquirer Assets	-5.27e-7	2.02e-7	-2.61^{***}	More: -
Acquirer Profit/Loss	0.4252	0.0814	5.23^{***}	Loss: +
Target Profit/Loss	0.1617	0.0793	2.04^{**}	Loss: +
Acquiring Country Accountability	-0.1886	0.0991	-1.90^{*}	More: +
Acquiring Country Corruption	0.2748	0.0817	3.36^{***}	More: +